

S1 File. Fitted receptive field models for the 31 verified first-order simple cells

Each row shows one cell. From left: (1) the raw ridge-regression map; (2) the same map smoothed at 0.75 pixel, as fitted, with the fitted first-order model's iso-response contours (solid: ON lobe; dashed: OFF lobe) and an arrow along the differentiation direction, pointing from the OFF to the ON lobe; (3-5) the idealised Gaussian derivative models of order 0, 1 and 2 fitted to the same map, with their R^2 .

The first-order model is the stored canonical fit; the zero- and second-order models are refitted with the same procedure and labelled with their recorded goodness of fit.

Angles are in degrees, counter-clockwise from rightward horizontal as displayed. ϕ is the differentiation direction of the fitted model on $[0^\circ, 360^\circ)$. σ is the geometric-mean scale, and $\kappa_{\text{dir}} = \sigma_{\perp}/\sigma_{\phi}$ (values below 1: elongated along the differentiation axis).

Intervals come from a parametric bootstrap: the cell's fitted model is re-embedded in noise taken from signal-free lags of its own reconstruction and refitted with the same procedure (100 draws). The bracket after κ_{dir} is the central 68% interval (16th to 84th percentile), and the value after ϕ is the 68th percentile of the orientation error. 68% intervals are shown because about 5% of refits converge to an alternative solution, which dominates the tails; 95% intervals are given in the accompanying table.

Cells are ordered by κ_{dir} , as in Fig 4.









